

RF and Microwave Engineering (EX752 / EX716)

Detailed PYQ

September 23, 2026

Notes

- Questions from Tribhuvan University, IOE (2069–2082 BS), 24 papers.
- Regular exam = **bold**, Back exam = regular font.
- BEX programme (EX752 / EG785) = normal font, BEI programme (EX716) = **this monospace style**.
- EX716 (BEI, IV/I) 7 papers 2079–2082; EX752 (BEX, IV/II) 16 papers 2070–2080; EG785EX (BEX Elective II) 1 paper, 2069 Bhadra.
- A header printing “Regular / Back” in one box counts as Regular, so **76 Bh**, **71 Bh** and **69 Bh** are bold.
- So: **72 Ash** = BEX Regular, 73 Ma = BEX Back, **79 Bh** = BEI Regular, 81 Ba = BEI Back.
- Months: Ba = Baishakh, Bh = Bhadra, Ash = Ashwin, Ma = Magh, Ch = Chaitra.
- Keywords: (I) = Importance/Need/Significance, (+) = Advantage, (-) = Disadvantage, +Fig = Figure, +Eg = Example, Vs = Compare/Differentiate.

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1 Introduction

(3 Hours)

1.1 Standard Frequency Bands & Microwave Applications

1. Classify microwave frequency bands + major applications [4] (69 Bh)
 - |→ How is the microwave frequency band classified by IEEE? [2+4] (80 Ba)
 - |→ Classify microwave frequency band + application in major areas [3+3] (79 Bh)
 - |→ How is microwave frequency band classified by the IEEE? (+) (-) of microwaves compared to lower frequencies [5+5] (76 Bh)
2. List areas of application of microwave systems [4] (70 Bh, 70 Ma)
 - |→ List uses of microwaves in communication systems + detailed protocols of any three [2+6] (81 Ba)
 - |→ Areas of application + detailed protocols of two microwave communication systems [3+3+4] (80 Bh)
 - |→ Real-world protocols of any three major applications of microwave communication systems [4+6] (82 Bh)
 - |→ Specifications/protocols of any five microwave communication systems [4+4] (82 Ba)
 - |→ Lists the areas of application of microwave system (with operational-principle comparison) [10] (78 Ch)

1.2 Behaviour: Conventional Vs RF/Microwave Circuits

1. Lumped Vs distributed circuit analysis; uses of microwave bands [4+2] (73 Ma)
 - |→ Differentiate lumped and distributed circuit analysis (+ protocols of two M/W comm systems) [3+3+4] (80 Bh)
2. Microwave Vs conventional low-freq systems: working principles [4] (70 Bh, 70 Ma)
 - |→ Compare behaviour of circuits for low-freq/conventional Vs RF/microwave bands [3+3] (79 Bh)
 - |→ (+) (-) of microwaves over acoustic waves [5] (72 Ash)
 - |→ (+) (-): microwave Vs conventional low-freq [6] (71 Ma)
 - |→ (+) (-) of microwaves compared to lower frequencies [2+4] (80 Ba)
 - |→ Circuit at seismic band Vs RF/Microwave counterparts [8] (75 Bh)
 - |→ Behavior of microwave passive components at low frequency bands Vs microwave bands [4+6] (82 Bh)
 - |→ Unique differences between microwave and conventional low frequency systems [4+4] (82 Ba)
 - |→ Distinguish microwave and traditional low frequency systems by operating principle; why S-parameter based analysis is a must [5+3] (81 Bh)
 - |→ How microwave systems are different from conventional low frequency devices [8] (80 Ch)
 - |→ Based on operational principles, compare microwave systems with conventional low frequency systems [10] (78 Ch)
 - |→ Behavior of microwave circuits; (+) (-) of using microwave frequency [4+4] (77 Ch)
 - |→ Merits and demerits of microwaves compared to seismic wave [5] (72 Ma)

2 RF and M/W Transmission Lines

(6 Hours)

2.1 Types of Transmission Lines

1. Microwave transmission lines Vs conventional low-freq lines [4] (69 Bh)
2. Differentiate the behavior of a transmission line at conventional low frequency and microwave bands [10] (79 Ch)
3. Microstrip Vs strip-lines [5] (71 Ma)

2.2 Smith Chart / Admittance (Immittance) Chart

1. Admittance chart [2] (72 Ash)
2. Immittance chart: sketch + list duality parameters [10] (75 Bh)
 - Sketch of an Immittance Chart (short note) [4] (81 Ba)
 - Sketch an immittance chart and compare the scales [5] (72 Ma)
 - Immittance Chart (short note) [5] (80 Ch)
3. Lossless 50Ω line terminated by $75+j100\Omega$; using Smith Chart find (a) Γ_L , (b) VSWR, (c) Z_{in} at 0.375λ from the load, (d) shortest length of line for which impedance is purely resistive, (e) the value of that resistance [10] (79 Ch)

2.3 Impedance Matching: Single Stub

1. Single stub shunt tuning (short & open): $Z_L = 80+j100\Omega$, microstrip [2+8] (72 Ash)
2. Single short/open shunt stub: $\Gamma_L = 0.5\angle 51^\circ$, $Z_0 = 50\Omega$ +Smith Chart [8] (74 Bh)
3. Single matching stub: 100Ω lossless line, complex load $120-j160\Omega$; all design steps with reasoning; S-Matrix before and after [5+3+2] (81 Bh)
4. Single short-circuited matching stub with self-defined line and load impedances, mentioning the steps [10] (79 Ch)
5. Single-stub tuner matching a lossless line to an antenna load, any assumed placement and length; derive its S-matrix [10] (76 Bh)
6. Signal shunt and open matching networks using Smith Chart: $Z_0 = 75\Omega$, $Z_L = 78.27+j60.93\Omega$; sketch the physical diagram considering microstrips [4+4+2] (72 Ma)

2.4 Impedance Matching: Double Stub

1. Double stub ($3\lambda/8$): $Z_L = 300+j300\Omega$, $Z_0 = 300\Omega$ +Fig [10] (71 Bh)
2. Double stub ($3\lambda/8$): $Z_L = 80+j180\Omega$, $Z_0 = 100\Omega$, 3 GHz +Fig [10] (70 Bh)
3. Double stub ($3\lambda/8$): $Z_L = 190+j110\Omega$, $Z_0 = 100\Omega$, 10 GHz +Fig [10] (70 Ma)
4. Double stub: inductive load, $Z_0 = 50\Omega$ +Fig [8] (73 Ma)
5. Double stub (short-circuit): $Z_L = 75+j75\Omega$, $Z_0 = 50\Omega$, $3\lambda/8$ [3+15] (69 Bh)
6. Double stub (short-circuit): complex inductive load, $Z_0 = 78.0\Omega$ +Fig [8+2] (73 Bh)
7. Double stub (short-circuit): $Z_L = 0.4+j0.85$ (normalized), $Z_0 = 75\Omega$ coaxial [10+2] (75 Bh)
8. Double stub shunt tuner (open stubs, $3\lambda/8$, first at 0.4λ): $Z_L = 60-j80\Omega$, 50Ω line; prepare S-matrix [8+2] (80 Ba)
9. Double stub (short-circuited coaxial): $Z_L = 109+j120\Omega$, 75Ω coaxial line; prepare S-matrix [8+2] (80 Bh)
10. Double stub: $\Gamma_L = 0.64\angle 58^\circ$, $Z_0 = 100\Omega$; prepare S-matrix of matched network [8+2] (81 Ba)

11. Double stub shunt tuner ($3\lambda/8$, first stub 0.01λ from load): $50\ \Omega$ lossless line, load admittance $0.00813 + j0.0065\ \text{U}$ +Smith Chart; write S-parameter [8+2] (**79 Bh**)
12. Broadband microstrip antenna, load reflection coefficient $0.33\angle 66^\circ$, transmission patch $75\ \Omega$; design the matching stubs and express the scattering matrix of the matched network [10] (**82 Bh**)
13. Double stub for an antenna at 10 GHz: $\Gamma = |0.45|\angle 60^\circ$, $100\ \Omega$ line; sketch physical diagram using micro strips; S-Matrices before and after [10+2+2] (**82 Ba**)
14. Double stub ($3\lambda/8$): $Z_L = 110 + j110\ \Omega$, $Z_0 = 100\ \Omega$; find lengths and spacing [12] (**80 Ch**)
15. Different impedance matching techniques; resulting S-matrix of a perfectly matched double-stub network [4+6] (**78 Ch**)
16. Sketch a double-stub perfectly matched network using microstrip and prepare its s-matrix [2+8] (**77 Ch**)
17. (+) of double stub matching over single stub matching; steps for impedance matching of a load using a double-stub network +Eg, using the provided Smith Chart [2+8] (**74 Ma**)

3 RF and M/W Network Theory and Analysis

(4 Hours)

3.1 S-parameters: (I), Definition, Properties

1. S-parameters (I) in microwave network analysis [4] (**70 Bh, 69 Bh**)
|→ S-parameters (I); Vs h-parameters for network analysis [4+4] (70 Ma)
|→ S-parameters (I) [4] (**75 Bh**)
2. S-parameters for two-port network [4+5] (**70 Bh**)
|→ S-parameters for two-port network [4+6] (**69 Bh**)
|→ S-parameters for two-port network [4] (70 Ma)
|→ Why are S-parameters used in microwave network analysis? Derive S-parameters for a two-port network [2+6] (80 Ba)
|→ Need of S-parameter + properties; three-port network parameters using E-plane Tee example [4+4] (**80 Bh**)
|→ Compare h-parameters with S-parameter; describe S-parameters using a two-port network [5+5] (**80 Ch**)
|→ Importance of S-parameters for microwave frequencies; evaluate the S-matrix of an Amplitude Modulator [3+7] (**78 Ch**)
3. S-parameters for three-port network [5+2] (**72 Ash**)
|→ S-matrix for 3-port network with +Eg [4+4] (73 Ma)
|→ Properties of 3-port network [4+4] (**73 Bh**)
|→ Three-port network model + its characteristic parameters [4] (**82 Bh**)
|→ Analyze a three port network using S-parameters [2+6] (74 Ma)
4. Return loss & insertion loss [2] (**72 Ash**)
5. Why are S-parameters used at high frequencies? For $[S] = \begin{bmatrix} 0.4+j0.5 & j0.6 \\ j0.6 & 0.4-j0.5 \end{bmatrix}$: (a) reciprocal? (b) lossless? (c) return loss at input? (d) if input power is 5 W, reflected power? [3+1+1+1+2] (**79 Bh**)
6. Find the basic differences of the systems having the S-Matrix sets (a) $S_{11}=0.65\angle 146^\circ$, $S_{12}=0.12\angle 46^\circ$, $S_{21}=2.30\angle 44^\circ$, $S_{22}=0.17\angle -172^\circ$ and (b) $S_{11}=0.30\angle -134^\circ$, $S_{12}=0.03\angle 46^\circ$, $S_{21}=2.20\angle 44^\circ$, $S_{22}=0.10\angle -172^\circ$ [4] (**82 Bh**)

4 RF/Microwave Components and Devices

(8 Hours)

4.1 Waveguides

1. Rectangular waveguide: TE mode field equations [7+3] (**70 Bh**)
 - |→ Rectangular waveguide: TM mode field equations + parameters [8+2] (**71 Bh**)
 - |→ Rectangular waveguide: TM mode field equations [7+3] (70 Ma) [10] (**74 Bh**)
 - |→ Rectangular waveguide: modes of propagation + critical parameters [8] (**69 Bh**)
 - |→ Field equations + basic characteristic equations of rectangular waveguide, TM_{11} mode [8] (81 Ba)
 - |→ TE_{mn} field equations of rectangular waveguide (short note) [4] (**80 Bh**)
 - |→ Express various field components of a rectangular waveguide in TM mode [8] (**81 Bh**)
 - |→ EM field equations and characteristic parameters of a rectangular waveguide working on TE mode [10] (**80 Ch**)
 - |→ Basic field equations of a rectangular waveguide working at TE mode [10] (**79 Ch**)
 - |→ Derive explicitly the field equations of rectangular waveguides for TE modes [8] (**77 Ch**)
 - |→ Field and characteristics equations of a rectangular wave guide working in TM mode [10] (**76 Bh**)
 - |→ Field equation of rectangular wave guide for TE mode (short note) [4] (74 Ma)
 - |→ TM mode rectangular wave guide (short note) [5] (72 Ma)
2. TE_{10} Vs TE_{20} : cutoff frequency + dominant mode [2] (**71 Bh**)
 - |→ Show 1 GHz cannot propagate in TE_{10} , 5 cm wall separation [3] (**70 Bh**)
 - |→ Prove TM_{01} and TM_{10} modes do not exist [3] (70 Ma)
 - |→ Prove that the lowest cut-off frequency mode in rectangular waveguides operating in TM mode is TM_{11} [3] (**82 Bh**)
 - |→ Show that the dominant mode in rectangular waveguides operating in TE mode is TE_{10} when $b > a$ [3] (**81 Bh**)
 - |→ Degenerative and dominant modes of a rectangular waveguide + Eg [3] (82 Ba)
 - |→ Dominant Vs degenerate modes; for $a = 3b$ find the dominant mode among TM_{01} , TM_{10} , TM_{11} , TM_{21} , TM_{12} , TM_{02} , TM_{20} [2+2] (72 Ma)
3. TE Vs TM mode: circular waveguide [10] (71 Ma)
 - |→ Fundamental field + characteristic equations of a circular waveguide, TE mode [10] (**79 Bh**)
 - |→ Electromagnetic field equations of a circular waveguide operating at TE Mode [7] (82 Ba)
4. Circular waveguide: TM wave field strength + dominant mode (TE & TM) [8+2] (**75 Bh**)
5. Dominant mode in waveguide [5] (**72 Ash**)
 - |→ What is dominant mode? Dominant modes for rectangular & circular waveguides. Does a rectangular waveguide of width 2.254 cm support 6 GHz? [1+1+2] (**80 Bh**)
6. For an air-filled rectangular waveguide (width 3 cm, $f = 6$ GHz, dominant mode): find cutoff frequency, cutoff wavelength, group velocity, phase velocity, guide wavelength and characteristic impedance. Also explain the four basic parameters of a directional coupler. [6+4] (80 Ba)
7. Rectangular waveguide with suitably assumed breadth, width and frequency (dominant mode): determine cut-off frequency, phase velocity, propagation wavelength in the waveguide and the characteristic impedance [6] (**77 Ch**)

4.2 Coupling: Probes and Loops

1. PROBE-coupling Vs LOOP-coupling [5] (**69 Bh**)

4.3 Tee Junctions

1. Waveguide junctions; Magic Tee operational principles (S-parameters) [3+3] **(72 Ash)**
 |→ Microwave magic tee [5] (71 Ma, **71 Bh**)
 |→ Draw a neat diagram of a magic tee and derive its S-parameter [4] **(80 Bh)**
 |→ Microwave Magic Tee (short note) [4] **(79 Bh)**
 |→ S-Matrix of a rectangular magic tee having both E and H-arms shorted, and how it behaves [4] **(82 Bh)**
 |→ Microwave magic tee (short note) [5] **(80 Ch)**
 |→ Magic Tee (short note) [5] **(79 Ch)**
 |→ Describe magic Tee based on S-parameters [6] (72 Ma)
2. E-plane tee Vs H-plane tee [5] **(69 Bh)**
 |→ E-plane Tee (short note) [4] (80 Ba)
3. Hybrid tee [8] (70 Ma)
4. Properties of a Hybrid-Tee, using proper S-Matrices, for (a) terminated E-arm, (b) terminated H-arm and (c) terminated co-planner arms [4+4+2] (82 Ba)
5. Properties of two Magic Tees with their E-arms connected; derive its S-matrix [10] **(77 Ch)**
6. Combinational effect of two magic tees connected in H-plane (short note) [5] **(78 Ch)**
7. 3-port directional coupler (S-parameters analysis) [4] **(75 Bh)**

4.4 Passive Microwave Devices: S-matrix Analysis

1. Identify + explain passive device from S-matrix (4-port) [8] **(74 Bh)**
2. Passive device from given S-matrix: judge condition + characteristics [6] **(75 Bh)**
3. Choose device to split power in half + explain properties [8] (73 Ma)
 |→ Two radar transmitters, double power to antenna: solution + Fig, math [8] **(73 Bh)**
4. Prepare S-Matrices of a perfectly working E-Plane Tee, H-Plane Tee and a Duplexer, and explore their uses [8] (81 Ba)
5. Identify and explain, with a neat diagram, the passive device described by the 4-port S-Matrix

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$
 [5] **(81 Bh)**
6. Identify and explain the properties of the passive device with S-Matrix

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$
 + Fig [10] **(78 Ch)**
7. Airport Surveillance Radar requiring half the input power to the antenna than the transmitter can deliver, using a duplexer; prepare its S-matrix [10] **(76 Bh)**
8. Two identical radar transmitters, twice the input power to an antenna; give the solution with explanation using S-matrix [6] (74 Ma)

4.5 Directional Couplers

1. Directional couplers [6] **(73 Bh)**
2. Directional coupler (short note) [5] **(79 Ch)**

4.6 Other Devices (Gunn diode, MASER)

1. Gunn-diode (short note) [4] (**79 Bh**)
2. MASER (short note) [4] (80 Ba)

4.7 Resonators and Circulators

1. Circulators [5] (**72 Ash, 70 Bh**, 73 Ma)
2. Microwave cavity resonators [6] (**74 Bh**)
3. Cavity Resonators (short note) [4] (74 Ma)

4.8 Microstrips

1. Microstrips [5] (**69 Bh**)

5 Microwave Generators

(5 Hours)

5.1 Transit-time Effect

1. Transit-time effect + two-cavity klystron construction & operation [2+8] (70 Ma)
2. What is transit time effect? Working principles of a magtron having 45° phase difference of adjacent cavities [2+6] (**77 Ch**)

5.2 Klystrons

1. Bunching effect (density modulation) [2] (**72 Ash, 71 Bh, 70 Bh**)
2. Bunching effect and field effect; working principle of a microwave high power oscillator employing both +Fig [2+2+4] (**81 Bh**)
3. Multi-cavity klystron oscillator working principle [2+7] (**72 Ash**) [2+8] (**71 Bh**)
 - |→ Two-cavity klystron [5] (**69 Bh**)
 - |→ Multi-cavity Klystron Amplifier (short note) [4] (**82 Ba**)
 - |→ Multi-cavity klystron (short note) [5] (**80 Ch**)
 - |→ Cross-field effect; construction and principle of operation of a multi cavity klystron amplifier [2+8] (**78 Ch**)
 - |→ Bunching effect in reflex klystron (short note) [5] (72 Ma)
4. Velocity modulation + bunching in multicavity reflex klystron +Fig [10] (**75 Bh**)

5.3 TWT and Magnetrons

1. Cavity magnetron: construction + operational features [2+8] (**70 Bh**) [10] (71 Ma)
 - |→ Sketch a cross-sectional magnetron with 45° phase shift among adjacent cavities; explain its functioning as a power amplifier + bunching effect [6+2] (**80 Ba**)
 - |→ Working principles of a cross-field magnetron with 60° phase differences among adjacent cavities [8] (**81 Ba**)
 - |→ Working principle of a circular beam-type microwave cavity-controlled cross-field amplifier [7] (**82 Bh**)
 - |→ Magnetron (short note) [5] (**79 Ch**)
 - |→ Working principle and applications of a cavity magnetron having 45° phase shift between the cavities [8] (74 Ma)
 - |→ Cross-field effect; design and working principle of a cross field cavity device for power amplifier, 90° phase shift between adjacent cavities [3+7] (72 Ma)
2. What is bunching effect? Describe the construction and operation of Traveling Wave Tube (TWT) [2+6] (**80 Bh**)
 - |→ How is the output of conventional tubes reduced at microwaves (inter-electrode capacitance, lead inductance, transit time)? Construction & working of TWT [2+6] (**79 Bh**)

5.4 Backward Wave Oscillator (BWO) and LNA

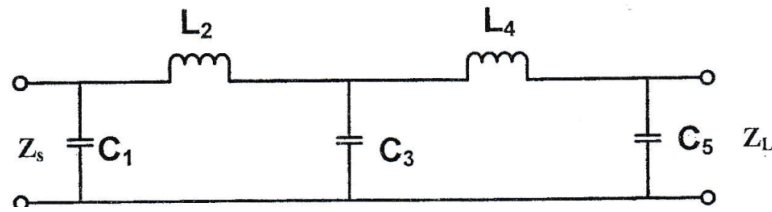
1. Bunching effect; working principle of BWO +Fig [2+8] (73 Ma)
 - |→ Backward Wave Oscillator (short note) [5] (**74 Bh, 71 Bh**)
2. Slow backward wave structure; LNA: construction + working principle [2+6] (**73 Bh**)

6 RF Design Practices

(10 Hours)

6.1 RF Filter: Insertion Loss Method & Microstrip Implementation

1. LNA cavity device – insertion loss method for filter design [5] (72 Ash)
2. Insertion loss method for filter design [5] (75 Bh) [8] (73 Bh)
 - |→ Butterworth & Chebyshev: common to two-port prototype filter (insertion loss) [4] (70 Bh, 70 Ma)
 - |→ Describe insertion loss method, illustrate passive LPF using μ -strip [8+2] (73 Ma)
3. Microwave filter design procedures [5] (71 Bh)
 - |→ Designing steps + passive HPF + Eg using microstrips [6+4] (74 Bh)
 - |→ Steps of microwave filter designing; sketch a double-pad T-type passive filter using microstrip [7+3] (81 Ba)
 - |→ Why filters are designed using the Insertion Loss Method; detail design steps of the filter below, implemented using microstrip [3+8+3] (82 Bh)



- |→ Microwave filter designing steps; sketch of a triple-pad series-arm type HPF using micro strips [7+3] (82 Ba)
- |→ Single section series arm passive BPF + Eg using microstrip; how filters are designed using the Insertion Loss Method [2+8] (81 Bh)
- |→ Double-section series-arm type micro strip filter designed using insertion loss method [10] (78 Ch)
- |→ Procedures for prototyping Butterworth LPF using insertion loss method [8] (77 Ch)
- |→ LPF prototype (Butterworth & Chebyshev) via insertion loss; implement a LPF double π -sections using microstrips [8+2] (76 Bh)
- |→ How is a LPF implemented using microstrip? How are LPFs prototyped? [5+3] (74 Ma)
- |→ Microwave filter parameters and LPF prototyping (short note) [5] (72 Ma)
- 4. LPF using microstrip + how prototyped [3+5] (72 Ash)
 - |→ Why is the insertion loss technique used to design microwave filters? Sketch a double-section shunt-arm microwave LPF using microstrips [3+5] (80 Ba)
 - |→ How is a LPF prototyped based on Butterworth/Chebyshev and converted to other filters? Implement a 2nd-order HPF π -section using microstrips [10] (80 Bh)
 - |→ How is a LPF prototype (Butterworth) designed using the insertion loss method? Implement a LPF π -section using microstrips [6+2] (79 Bh)

6.2 RF Amplifier: Theory, Stability, Design

1. Amplifier Vs oscillator circuit: stability factor [5] (71 Ma)
2. Stability + max gain (conditional stable, unilateral): $S_{11}=0.894\angle-60.6^\circ$, $S_{12}=0.020\angle62.4^\circ$, $S_{21}=3.122\angle123.6^\circ$, $S_{22}=0.781\angle-27.6^\circ$ [5+5] (72 Ash)
 - |→ Same S-parameters: justify conditionally stable [10] (70 Ma)
 - |→ Same S-parameters (at 6 GHz): investigate the stability [5+5] (80 Ba)
 - |→ Same S-parameters: determine stability + compare max power gains for bilateral & unilateral

- modes [5+5] **(79 Bh)**
3. Stability + max gain (bilateral + unilateral): $S_{11}=0.55\angle 150^\circ$, $S_{12}=0.04\angle 20^\circ$, $S_{21}=2.82\angle 180^\circ$, $S_{22}=0.45\angle -30^\circ$ [10] **(71 Bh)**
 |→ Max gains of both bilateral & unilateral models; sketch flowchart of microwave amplifier designing procedure [6+3+3] **(81 Ba)**
 |→ Calculate and compare max power gain for unilateral & bilateral modes, with $S_{11}=0.55\angle -150^\circ$ as printed [15] **(71 Ma)**
 4. Stability + max gain (bilateral + unilateral + Γ_{in} , Γ_{out} on Smith Chart): $S_{11}=0.656\angle 146.7^\circ$, $S_{12}=0.122\angle 46.1^\circ$, $S_{21}=2.30\angle 44.7^\circ$, $S_{22}=0.172\angle -117.1^\circ$ [4+4+4+4] **(74 Bh)**
 |→ Max gain (bilateral + unilateral): same S-parameters [10] **(70 Bh)**
 5. GaAs FET at 4 GHz: $S_{11}=0.72\angle -116^\circ$, $S_{12}=0.03\angle 57^\circ$, $S_{21}=2.60\angle 76^\circ$, $S_{22}=0.73\angle -54^\circ$ [18] **(69 Bh)**
 6. Stability + compare max power gains for unilateral and bilateral modes using the set (a) above: $S_{11}=0.65\angle 146^\circ$, $S_{12}=0.12\angle 46^\circ$, $S_{21}=2.30\angle 44^\circ$, $S_{22}=0.17\angle -172^\circ$ [12] **(82 Bh)**
 7. Design a microwave amplifier for maximum power gain, bilateral and unilateral: $S_{11}=0.55\angle -150^\circ$, $S_{12}=0.04\angle 20^\circ$, $S_{21}=2.82\angle 180^\circ$, $S_{22}=0.45\angle -30^\circ$ [8+4] **(82 Ba)**
 8. Stability + unilateral gain: $S_{11}=0.45\angle -160^\circ$, $S_{12}=0.04\angle 20^\circ$, $S_{21}=3.12\angle 180^\circ$, $S_{22}=0.35\angle -40^\circ$ [7+8] **(80 Ch)**
 9. GaAs power FET at 5 GHz, 50 Ω : $S_{11}=0.45\angle 163^\circ$, $S_{12}=0.04\angle 40^\circ$, $S_{21}=2.55\angle -106^\circ$, $S_{22}=0.46\angle -65^\circ$; design an amplifier and find the total power gain [10] **(79 Ch)**
 10. GaAs FET at 5 GHz, 50 Ω : same S-parameters; check stability and find the maximum power gain [10] **(72 Ma)**
 11. Ku-Band satellite transmit amplifier, High Electron Mobility GaAs MOSFET at 4 GHz, 50 Ω : $S_{11}=0.72\angle -11^\circ$, $S_{12}=0.03\angle 57^\circ$, $S_{21}=2.60\angle 76^\circ$, $S_{22}=0.73\angle -54^\circ$; stability + compare max power gain, bilateral and unilateral [5+5] **(77 Ch)**
 12. Microwave transistor at 1.0 GHz, 50 Ω : $S_{11}=0.38\angle -158^\circ$, $S_{12}=0.11\angle 54^\circ$, $S_{21}=3.50\angle 80^\circ$, $S_{22}=0.40\angle -43^\circ$, with $Z_s = 25\ \Omega$ and $Z_L = 40\ \Omega$; compute the gains of input and output matching networks and draw a design flowchart [6+4] **(76 Bh)**
 13. Stability + compare max power gain, unilateral and bilateral: $S_{11}=0.78\angle -113^\circ$, $S_{12}=0.028\angle 247^\circ$, $S_{21}=2.60\angle 76^\circ$, $S_{22}=0.81\angle -54^\circ$ [12] **(74 Ma)**
 14. GaAs MESFET at 10 GHz: $S_{11}=0.55\angle -160^\circ$, $S_{12}=0.04\angle 10^\circ$, $S_{21}=4.82\angle 180^\circ$, $S_{22}=0.45\angle -30^\circ$; determine stability + compare max power gains for unilateral & bilateral [10] **(80 Bh)**
 15. Stability + max gain (bilateral + unilateral): $S_{11}=0.64\angle -169^\circ$, $S_{12}=0.03\angle 50^\circ$, $S_{21}=10.11\angle 91^\circ$, $S_{22}=0.22\angle -82^\circ$ [12] **(73 Ma)**
 16. Flowchart for microwave amplifier design (GaAsFET) [4] **(74 Bh)**
 |→ Flowchart + stability of $C_S=1.15\angle 10^\circ$, $R_S=0.85$, $C_L=1.10\angle 80^\circ$, $R_L=1.10$ [5+5] **(71 Bh)**
 17. Stability synthesis from Smith Chart (conditional stability) [8] **(74 Bh)** [10] **(73 Bh)** [5] **(75 Bh)**
 |→ Stability analysis of a microwave amplifier (short note) [4] **(81 Ba)**
 |→ Microwave Amplifier Design Flow and Stability Analysis (short note) [4] **(82 Ba)**
 |→ Investigate the stability with a neat flow diagram: $[S] = \begin{bmatrix} 0.6\angle -140^\circ & 0.03\angle 62^\circ \\ 2.40\angle 54^\circ & 0.70\angle -58^\circ \end{bmatrix}$ [3+8] **(81 Bh)**
 |→ Max gain of an amplifier in a unilateral model with assumed S-parameters; flow chart of the designing steps [10] **(78 Ch)**

6.3 Oscillator and Mixer Theory

1. Mixer theory [5] **(70 Bh)**
 |→ Microwave mixer (short note) [5] **(72 Ma)**
2. Microwave oscillator theory [8] **(70 Ma)**

7 Microwave Antennas and Propagation

(3 Hours)

7.1 RF/Microwave Radiation, Safety Practices and Standards

1. Microwave radiation hazards (radiation fields) [8] (**69 Bh**) [5] (**70 Bh**, 73 Ma)
 - |→ Microwave radiation hazards + safety practices (short note) [5] (**71 Bh**)
 - |→ RF radiation hazards + safety standards (short note) [5] (70 Ma)
 - |→ RF/MW radiation hazards + safety practices [6] (**73 Bh**, **74 Bh**) [3+3] (**79 Bh**)
 - |→ Effect of SAR as microwave radiation hazard [6] (**74 Bh**)
 - |→ How microwave radiation becomes hazardous to human body [5] (**75 Bh**)
 - |→ Discuss RF radiation fields to public exposers; international standards and recommended (SARPs) practices to be safe [3+3] (80 Ba)
 - |→ Classify microwave propagation fields of a GSM BTS + radiation parameters; what non-ionizing SAR is and how it becomes hazardous [4+4] (81 Ba)
 - |→ HERP (short note) [4] (**80 Bh**)
 - |→ How microwaves are hazardous to humans [4] (**82 Bh**)
 - |→ Compare radiation fields around a mobile BTS at 900 MHz and 1800 MHz with their quarter-wavelength antennas; how microwave EMR becomes hazardous; its IEEE safety standard [4+2+2] (82 Ba)
 - |→ Approximate the hazardous and safe zone for a BTS tower at 2600 MHz with antenna size 0.5 m; how non-radiating radiation is hazardous to the human body [4+4] (**81 Bh**)
 - |→ Define SAR; microwave radiation hazards based on SAR exposures [10] (**80 Ch**)
 - |→ Five major RF safety practices; differentiate microwave reactive, near- and far-field radiations + Fig [5+5] (**79 Ch**)
 - |→ Microwave radiation hazards (short note) [5] (**78 Ch**)
 - |→ How are microwaves hazardous to humans? Define the different radiation zones of a microwave oven [3+3] (**77 Ch**)
 - |→ Name and explain different microwave radiation fields; different international EMR safety standards and practices [5+5] (**76 Bh**)
 - |→ Types of electromagnetic radiation hazard; illustrate the radiation fields of a microwave oven [4+4] (74 Ma)
 - |→ Compare RF radiation fields and explain how they are hazardous to the living body [5] (72 Ma)
2. Microwave radiation fields [5] (71 Ma)
3. Antenna as DUT: choose measurement tool + working principle [8] (**74 Bh**)

8 RF/Microwave Measurements

(6 Hours)

8.1 Power Measurement: VSWR and Bolometry

1. Standing waves + microwave power: VSWR meter + low power measurement [2+8] (**72 Ash**)
|→ VSWR meter + bolometry [4+6] (**70 Bh**)
|→ List major RF/MW measurement parameters; how is the VSWR of a microwave transmitter (VSWR > 10) measured? [2+6] (**79 Bh**)
2. Microwave measurements Vs low-frequency measurements [3] (**71 Bh**)
3. Major microwave measurement parameters + low power measurement device [8] (**73 Bh**)
4. Power measurement using calorimeter wattmeter; limitations of a single-bridge bolometer [6+2] (**80 Ba**)
|→ Power measurement + working principle of the double-channel Bolometer Bridge method [8] (**80 Bh**)
|→ Double-channel Bolometer (short note) [4] (**81 Ba**)
5. Spectrum Analyzer (short note) [4] (**80 Bh**)
6. Working principle of a microwave measuring device to map different radiation zones around a GSM BTS [8] (**82 Bh**)
7. Working principle of bolometry used in microwave measurements [10] (**79 Ch**)
8. Bolometry (short note) [5] (**80 Ch**)
9. Choose a proper power measurement tool to measure the power of an Airport Surveillance Radar [8] (**74 Ma**)
10. Measuring about 7.5 mW of microwave power: choose a proper device and explain its working principle [10] (**72 Ma**)
11. Spectrum Analyzer (short note) [5] (**80 Ch**)

8.2 Calorimeter Methods

1. Static calorimeter working principle [5] (**71 Ma**) [7] (**71 Bh**)
2. Calorimetry in microwave; circulating Vs flow calorimetry [2+8] (**70 Ma**)
|→ Circulating calorimeter (power measurement) [5] (**75 Bh**)
|→ Circulating Calorimetry (short note) [4] (**82 Ba**)
|→ Microwave medium power measured with static and dynamic circulating calorimetry; list all microwave measurables and their measuring tools [6+3] (**81 Bh**)
|→ How microwave power is measured with a static calorimeter [6] (**77 Ch**)
|→ Power measurement using static and dry calorimeters [10] (**76 Bh**)

8.3 Network Analyzer and Two-Port Models

1. Two-port network model + derive parameters [10] (**71 Bh**)
2. Network analyzer working principle [8] (**73 Ma**)
3. VNA type Network analyzer (short note) [5] (**78 Ch**)